

VIBHAV MISRA

New York, NY 11206 | vibhavemail@gmail.com | [LinkedIn](#) | [GitHub](#) | [Website](#)

Work Authorization: Eligible to work in the U.S. for 3 years (F-1 OPT/STEM Extension). No sponsorship required.

SUMMARY

Data professional with an M.S. in Data Science and experience across analytics, machine learning and business intelligence. Skilled in Python, SQL, Power BI, Tableau, and statistical modeling, with hands-on experience building predictive models, dashboards, forecasting workflows, and KPI-driven reporting solutions. Strong ability to transform raw data into actionable insights, evaluate model performance, and communicate findings for business and operational decision-making.

EDUCATION

Pace University, Seidenberg School of Computer Science and Information Systems	New York City, NY
Master of Science (MS) in Data Science Honors: magna cum laude	May 2026
Chandigarh University, Apex Institute of Technology	India
Bachelor of Engineering (BE) in Computer Science with a specialization in Artificial Intelligence and Machine Learning	June 2024

WORK EXPERIENCE

EdMyst Inc.	Lewes, DE
Data Science Intern	June 2025 - September 2025
- Designed long-term memory architecture for an AI coaching platform using user profiles, session reports, feedback, and behavioral metrics to enable personalized coaching across sessions. - Built a Python context builder module to structure user memory, feedback, reports, metrics, and profile data for AI prompt generation. - Developed technical specifications for integrating FastAPI, MongoDB, AWS S3, AWS Lambda, and OpenAI APIs into a personalized AI Coach backend. - Automated Google Sheets reporting workflows using Apps Script to consolidate candidate data into a centralized dashboard with KPI, trend, client-level, and error-analysis views.	

PROJECTS

Global Economic Stress Early-Warning System	May 2026
- Addressed severe class imbalance in macroeconomic crisis data by implementing and comparing deep generative models (CTGAN, TVAE) and Gaussian Copulas to synthesize high-fidelity minority class events. - Trained and optimized XGBoost, Random Forest, and Logistic Regression models, engineering specialized features such as Yield Curve Inversion tracking and a custom Economic Stress Index (ESI). - Structured chronological cross-validation to eliminate look-ahead bias, culminating in an optimized XGBoost-Jittering ensemble that achieved 94.6% accuracy and a peak 0.788 Recall for risk mitigation.	
NYC 311 Service Quality & Operational Performance Dashboard	February 2026
- Analyzed 5M+ service requests using a PostgreSQL star schema to evaluate agency performance and SLA adherence across NYC boroughs. - Developed an interactive Power BI dashboard with "as-of-date" logic to track volume, resolution efficiency, and backlog trends for 15+ agencies. - Identified critical bottlenecks (e.g., 3x city-median resolution times) and provided data-driven recommendations to improve operational efficiency.	
Customer Review Analytics & Booking Prediction - British Airways Case Study (Forage)	November 2024
- Cleaned and analyzed 1,000+ airline customer reviews to identify key service drivers impacting booking completion. - Engineered structured features and performed exploratory data analysis to uncover themes related to delays, service quality, and amenities. - Developed a predictive model to highlight factors influencing booking behavior, translated findings into business recommendations. - Presented insights in a stakeholder-style summary report outlining actionable service improvement areas.	

TECHNICAL SKILLS

Languages & Analytics: Python, SQL, R, SAS, Excel
ML & Data Science: Pandas, NumPy, Scikit-learn, Keras, Feature Engineering, Classification, Regression, Forecasting, Model Evaluation
BI & Visualization: Power BI, Tableau, Looker, Matplotlib, Seaborn, Plotly, KPI Reporting, Dashboard Development
Databases & Tools: PostgreSQL, MySQL, MongoDB, Star Schema Design, ETL, AWS S3, AWS Lambda, Git/GitHub, Streamlit, FastAPI, Flask

ADDITIONAL COURSEWORK

Indian Institute of Technology (IIT) Madras <i>Foundational Certificate, Data Science</i>
Core Coursework: Machine Learning (Foundations & Techniques), Business Data Management, and Business Analytics.